Charles Herman

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Summary

Mechanical engineering student with hands-on experience in rocketry, mechanical design, and technical presentation. Contributed to hybrid propulsion development and recovery systems, and recently assumed the role of Chief Engineer within The University of Melbourne's high-powered rocket team, ARES. Known for practical problem-solving, reliability, and effective collaboration across technical groups. Recently applied these skills as an Undergraduate Engineer at Toyota Motor Corporation Australia, building on previous engineering consulting experience.

Education

- **University of Melbourne**, Bachelor of Science (Mechanical Engineering Systems) *Expected 2026*
- **Wesley College**, International Baccalaureate (IB) Diploma *2020 – 2021*

Experience

- **Undergraduate Engineer – Toyota Motor Corporation Australia** *Nov 2025 – March 2026*
  Working within the Product Planning and Development division to support mechanical design and operational efficiency. Responsibilities include assisting with part fitment, CAD modelling using CATIA, conducting field testing and FEA validation, and performing data analysis to support design optimisation.
- **ARES Rocketry Team – Chief Engineer** *July 2025 – Present*
  Leading the technical development of high-powered experimental rockets, coordinating multidisciplinary teams and overseeing system integration. Responsibilities include ensuring design robustness, meeting competition requirements, and guiding component testing and validation.
- **ARES Rocketry Team – Propulsion/ Recovery Sub-Team Member** *Mar 2024 – July 2025*
  Completed LDR, PCR and CDR, and fabricated a spin-cast system for paraffin fuel grain manufacturing. Developed MATLAB-based hybrid motor sizing scripts to determine appropriate combustion chamber dimensions. Built and tested parachute deployment systems, delivered technical presentations at the Victorian Rocketry Association, and co-authored technical documentation for competition judges.
- **Undergraduate Engineer – Ascot Consulting Engineers** *July 2025 – Nov 2025*
  Supporting senior engineers on building services projects, learning across multiple disciplines, attending client meetings, and developing practical skills in AutoCAD and Revit.
- **Regioneering Volunteer – Engineers Without Borders** *Nov 2024 – Dec 2024*
  Participated in an outreach trip to rural Victorian primary schools aimed at increasing STEM engagement.

Full work history and references available upon request.

Skills

- **Software:** SolidWorks, Onshape, MATLAB, Dewesoft, CREO, Fusion360, 3D Experience.
- **Fabrication:** Soldering, Metalworking, 3D Printing, Rapid Prototyping, Vehicle assembly.
- **Skills:** DFM and CNC, GDandT, Strain gauging, Model analysis, technical report writing, team leadership, adaptability.